

Music-Gen — Cycles 7-9

2026-08-28

Contents

Music-Gen — Cycles 7-9	1
Abstract	1
Introduction	1
Approach	2
Findings	3
Discussion	6
Open Questions	7
Appendix: Provenance	8

Music-Gen — Cycles 7-9

Abstract

Cycles 7-9 turned the campaign’s rules-and-score axis from partially-built into fully-actionable and closed the outstanding infrastructure hole for rated-audio acquisition. Cycle 7 was the post-merge integration pass for fork 22b8c654f616 (three sibling clones landing the source-separation and heuristics workstreams). Cycle 8 ran fork 3a908edcb241 — three parallel clones delivering M-SCORE-1 (the MuseScore programmatic bridge), the M-TRANS-1 basic-pitch octave-suppression negative-finding sub-milestone, and M-INGEST-1/egress-ready-automation (the crash-resumable state machine that closes the “egress opens mid-run and nobody catches it” hole) — with clone 2 correctly using a second termination cycle to emit COMPLETE after cycle 1’s VALIDATED/high. Cycle 9 ran fork f1bae241bde9 — two clones delivering the M-RULES-1 extraction-half (28 typed rule rows across five rule types on the frozen 30 s merged score, byte-deterministic and provenance-resolvable) and the M-TEX-1 stage-by-stage measurement (24 panel numbers across three ordered audio stages, deterministic byte-for-byte, with an honest family-disagreement finding that validates the panel’s refuse-to-aggregate contract). At cycle-9 exit, both halves of M-RULES-1 (schema + extraction) and both halves of M-TEX-1 (panel + stage-by-stage) are validated; M-SCORE-1 is validated/high; and the campaign’s rules-ledger side of Goal G4 can now proceed independently of the rated-audio blocker. The egress-ready state machine is on disk, IDLE, and awaits two consecutive fresh media_ok=true rows to fire the rated-audio pipeline unattended.

Introduction

At the start of cycle 7 the campaign had the ingestion chassis, classifier, DAW spike, heuristics battery, texture panel, source separation, and transcription survey in place, plus the schema-half of M-RULES-1, an ear-model preparation chassis, and the MuseScore programmatic bridge specified but not built. The eight-song rated corpus was still blocked by an egress denial of *.googlevideo.com. The three cycles reported here were driven by three questions: can the merged score be produced deterministically so downstream rules extraction is unblocked; can the “nobody notices when egress opens” hole be closed programmatically

rather than by human polling; and can the two remaining halves of M-RULES-1 and M-TEX-1 be closed on the synthetic corpus so the rules-ledger side of Goal G4 is complete without waiting on rated audio. The answer to all three is yes.

Approach

Cycle 7 (post-merge integration, fork 22b8c654f616). A worker-only cycle folded three prior clones (M-SEP-1 htdemucs adoption, M-HEUR-1 heuristics battery, M-TRANS-1 transcription survey) into the main workspace. The pattern was the standard one from earlier cycles: propagate sub-milestone identifiers into the five-column plan-of-record table, adopt orphan artefacts via `_infra/adopt-fanout-artifacts-*` ledger events, and re-run the cross-branch integration test at the merged state. This is prior context for what follows, not a load-bearing contribution.

Cycle 8 (fork 3a908edcb241, three clones). Three parallel branches ran with disjoint file trees:

- Clone 0 built M-SCORE-1: `scripts/score/{bridge, jsonl_to_midi, seed_score}.py`, driving `mscore3 3.2.3` headless under `QT_QPA_PLATFORM=offscreen`, with a public API of `xml_to_midi`, `midi_to_xml`, `merge_stems_to_score`, and `ScoreBridgeError`. Two non-obvious mechanics were surfaced empirically and folded in: a determinism-scrubbing list for `mscore3`'s time- and build-varying metadata (`<encoding-date>`, `<software>`, `<source>`, `<encoder>`, `<supports>`, `mscore`'s default `<creator>`), plus a first-occurrence renumbering of the 32-hex-digit ids `music21` assigns to `<part>` / `<score-part>` / `<score-instrument>` / `<midi-instrument>`; and an interval-graph-coloring workaround for `mscore3`'s per-part MIDI voice-cap (six voices in one `<part>` collapse to fewer note-on streams on export), splitting into `{stem}__v{k}` parts with a `parts_mapping.json` sidecar. `music21 9.1.0` was pinned at the top level for this branch.
- Clone 1 built the M-TRANS-1/basic-pitch/octave-suppression sub-milestone. The filter groups notes into 25 ms co-onset buckets, enumerates within-bucket pairs whose pitches differ by exactly 12 semitones, qualifies pairs by $\text{dur}_{\min} \geq T_{\min_ms}$ and $\text{overlap_frac} \geq \text{overlap}_{\min}$, and iterates qualified pairs in confidence-descending order with velocity $\rightarrow$ duration $\rightarrow$ lower-pitch tie-breaking; on tie the higher-pitched member is the loser, preserving the bass fundamental. The pass is single-pass by contract. A 3×3 grid over $T_{\min} \in \{50, 100, 200\}$ ms and $\text{overlap}_{\min} \in \{0.3, 0.5, 0.7\}$ was published as a nine-cell heatmap co-located with the TSV.
- Clone 2 built M-INGEST-1/egress-ready-automation as a crash-resumable state machine over `IDLE  $\rightarrow$  ARMED  $\rightarrow$  TRIGGERED  $\rightarrow$  HARVESTING  $\rightarrow$  CHUNKING  $\rightarrow$  CLASSIFYING  $\rightarrow$  READY` with a `FAILED` sink. The trigger rule — two consecutive fresh `media_ok=true` rows in the probe log with a strict 24-hour staleness filter — is deliberately not tunable through the CLI. `state.json` is written atomically per transition (`NamedTemporaryFile + fsync + os.replace`); `transitions.jsonl` is append-only. `subprocess.run` is replaced at import time with `_SubprocessRunForbidden` in tests, so no real network was exercised.

The three branches merged into the main workspace with no file-tree overlap and no environment conflict; `music21 9.1.0` was added top-level without breaking the `numpy 1.26.4` / `TensorFlow 2.21` / `basic-pitch-in-venv` stacks. Clone 2 correctly used a second termination cycle to emit `COMPLETE` after its cycle-1 `VALIDATED/high` — a null cycle on genuinely-exhausted scope, per the pattern the campaign has now established.

Cycle 9 (fork f1bae241bde9, two clones). Both branches consumed cycle 8's M-SCORE-1 output as their common upstream:

- Clone 0 built the M-RULES-1 extraction-half. Five per-rule-type extractors (harmonic-

v1, rhythmic-v1, melodic-v1, form-v1, arrangement-v1) read the frozen data/score/merged_synth030s.musicxml via music21 9.1.0 and emit typed rule rows through the M-RULES-1/schema/ledger-writer. rule_id is derived by the pre-existing derive_rule_id(rule_content) helper — a SHA-256 over the canonical JSON of {rule_type, scope, sorted provenance_pointers, parameters}, never ts/extractor/event_id. event_id derives deterministically from rule_id; ts is pinned at a constant so re-runs are byte-identical.

- Clone 1 built the M-TEX-1/stage-by-stage measurement. Three ordered audio stages at 44.1 kHz stereo: original.wav (a byte-stable rewrite of the M-SEP-1 synth_030s/mix.wav via scipy.io.wavfile, dodging libsndfile’s per-write creation-date chunk), bare_midi.wav (fluidsynth on the M-SCORE-1 merged MIDI with argv byte-identical to scripts/separation/synth_gt.py and an asserted SF2 SHA), and effects_layered.wav (a pinned DawDreamer chain — Surge XT Chorus + Reverb + post-hoc gain envelope — with determinism pins applied before any DawDreamer import). The frozen 8-key texture panel was run across all three ordered pairs (24 numbers), and the report refuses to aggregate.

Findings

M-SCORE-1 (cycle 8, clone 0) — validated/high

The 8-bar seed round-trips to byte identity across two full xml → mid → xml → mid → xml passes after scrubbing; note preservation is 88/88 events with 0.00 ms onset drift. On the 30 s M-SEP-1 synth mix the merged full-song identity-merge F1 is **1.0000 / 1.0000 / 1.0000** on drums / bass / other against the basic-pitch input MIDIs — the bridge preserves every input note. F1 vs the M-SEP-1 tiled ground truth is 0.0000 / 0.4746 / 0.7317, upper-bounded by cycle-6 basic-pitch quality: the drums row is the lower-bound zero-notes case (basic-pitch emits no notes on a pitchless drum stem); the bass row reproduces docs/transcription_survey_report.md §5 bit-for-bit; the other row raises the cycle-6 baseline of 0.72 slightly because every input note survives the merge and recall reaches 1.0. The brief’s explicit relaxation clause for basic-pitch-upstream-bounded shortfalls applies, and the report §3 diagnosis is thorough. All four legibility contracts hold: malformed MusicXML (mscore3 returns exit 0 on structurally-invalid input and writes an empty MIDI — the bridge scans stderr for "is not a valid musicxml file", "is not a musicxml score-partwise file", "cannot import", "empty score" and escalates), missing input, timeout, and missing per-stem MIDI each surface as ScoreBridgeError with a useful diagnostic. Merge snaps every input onset and duration to a 1/64-quarter grid (~7.8 ms at 120 BPM; ≤ 3.9 ms max shift, well under mir_eval’s 50 ms tolerance) because music21’s serializer refuses arbitrary sub-tuplet durations. Test suite is 23/23 green.

M-TRANS-1/basic-pitch/octave-suppression (cycle 8, clone 1) — invalidated/high (negative finding, escape hatch)

The 3×3 grid across T_{min} × overlap_{min} publishes:

T _{min} overlap _{min}	0.3	0.5	0.7
50 ms	+0.1513	+0.1513	+0.1513
100 ms	+0.1513	+0.1513	+0.1513
200 ms	+0.1152	+0.1152	+0.1152

Best-cell aggregate bass F1 uplift is **+0.1513** (baseline 0.4773 → 0.6286); the **+0.3 success bar is not met on any cell**. The overlap_{min} axis is flat (the overlap_frac distribution on the

cycle-6 bass JSONL is bimodal near 1.0 for real octave-doubling artefacts and near 0.0 for spurious non-artefact pairs, and none of the swept values lands between the modes); $\tau_{\min}$ has a single step at 200 ms; the two axes collapse to one useful trust-threshold knob in [50, 100] ms. Diagnostic on `synth_030s/bass` shows the filter correctly suppresses the (33 → 45) and (36 → 48) octave pairs (14 notes removed, precision jumped 0.318 → 0.467), but fails to suppress the chained (45 → 57) and (48 → 60) pairs because the specified single-pass rule skips any pair whose either member is already suppressed — exactly the “chain of three octaves” edge case the brief’s mechanism section anticipated as a known limitation. The harmless-to-others constraint is trivially satisfied everywhere (drums Δ = other Δ = 0.0000, mechanical from the bass-only driver). The escape hatch was invoked cleanly: no re-tuning of the co-onset window, the trust-threshold definition, or the harmless-to-others constraint. The audit reproduced the TSV SHA-256 live (d87fa0f5e6d87e6be551fcfb4e844a35c247733c42b19452d416b5ba573b0ec2) and concurred with the recommendation to reopen only via fixed-point iteration in a later cycle (one-line wrapper, expected +0.10 additional aggregate F1 based on the pitch-set arithmetic).

M-INGEST-1/egress-ready-automation (cycle 8, clone 2) — validated/high, then COMPLETE

Six named scenarios pass end-to-end against synthetic fixtures at `tests/fixtures/egress_status/` with the reference clock frozen to 2026-08-28T10:00:00Z: `all-false` → IDLE; `single-true-then-back` → IDLE; `two-consecutive-triggers` → TRIGGERED(1, 2) → READY; `already-triggered-then-false` → READY (state authoritative; re-scan does not retract); `interleaved-then-true-true` → TRIGGERED(2, 3) → READY; `stale-row-does-not-count` → ARMED((1,)) because the 24 h-old row is invisible to the fresh-only filter. Every drive-through additionally asserts that `state.json` on disk matches the terminal in-memory state (or is absent when the machine never left IDLE), that `transitions.jsonl` records the correct sequence and is append-only, and that a second `EgressReadyMachine(...)` against the same disk state returns the same terminal state without re-firing any hook (idempotence). Byte-deterministic `transitions.jsonl` across two `--watch` invocations against the same fixture (SHA-256 equal); atomic `state.json` under a monkey-patched `os.replace` mid-transition (previous bytes remain readable); zero live `subprocess.run` calls; zero `sidecar_nonfactor` imports. `--resume` restarts only the failed stage and its successors; `--reset-failure` requires `--force-idle` as a two-flag acknowledgement, refused otherwise with CLI exit 2. Test suite 62/62 green; §17 of the cross-branch integration test adds 52 checks and is green. Cycle 2 was a null cycle on exhausted scope, emitting COMPLETE with [[BRANCH_COMPLETE]]; the runtime state artefacts (`state.json`, `transitions.jsonl`) are correctly absent until the first live trigger fires.

M-RULES-1/extraction (cycle 9, clone 0) — validated/high; parent M-RULES-1 closed

Twenty-eight rules land on `data/rules/ledger.jsonl` at ledger SHA-256 4fe722adde034c099ff9e65437f0d5c138cb3dd2

Rule type	Rows	Threshold	Representative content
harmonic	6	≥ 5	key=F_major, chord_ progression=[V, vii, iii, I, i, I, II, ii], cadence=none

Rule type	Rows	Threshold	Representative content
rhythmic	6	≥ 5	tempo_bpm=120.0, meter=4/4, pattern=[...32 cells...], swing_ratio=0.5
melodic	6	≥ 5	contour=static, range_ semitones=24, PCH sum=1.00000000
form	5	≥ 5	five sectionizations (monolithic, uniform-4m, uniform-2m, ABAB-4m, A-B-A halves)
arrangement	5	≥ 5	instrumentation=[drums,bass,othe density_over_ time=[...], layer_events=[...]
total	28	≥ 25	5.6× margin on aggregate

Every row passes `validate_batch()` (28 / 28, zero errors); two independent runs produce byte-identical ledgers; every provenance pointer re-hashes to its declared source (28 / 28); `read_ledger()` returns rows in first-seen [harmonic, rhythmic, melodic, form, arrangement] order; `effective_rules()` (no supersedes this cycle) equals `read_ledger()` (28 == 28); the 34-assertion extraction test suite is green; the 25-assertion schema regression suite is green; the AST scan for `sidecar_nonfactor` across `scripts/rules/extract/` returns zero hits. Three honest limitations are documented and disclosed rather than papered over: the rhythmic extractor falls back to bass onsets because the frozen basic-pitch drums stem is empty (every hit is labelled "kick" as a onset-grid placeholder; a future `rhythmic-v2` supersedes when a real drums transcription lands); the form extractor emits five parallel sectionizations of the seed rather than five detected sections (a grammar-fit workaround because `form.parameters` only accepts a sections list — a `form-v2` grounded on novelty-curve boundaries is the right supersedes); the merged score reports 131 nominal measures for a 30 s clip because `music21` sees trailing empty measures across the ten sub-parts (extractors correctly honor the nominal count; upstream M-SCORE-1 refinement flagged). None of the three is a defect blocking the parent milestone. Both halves of M-RULES-1 (schema, cycle 6; extraction, this branch) are now closed.

M-TEX-1/stage-by-stage (cycle 9, clone 1) — validated/medium; parent M-TEX-1 validated/medium

The 24 texture-panel numbers on the three ordered pairs:

a_stage	b_stage	mel_l1_db	spectral_ centroid_ rmse_hz	rms_env_ rmse	lufs_m_ rmse_lu	embedding_ cosine_ distance
original	bare_ midi	9.906	2804.9	0.0276	2.682	0.1234
original	effects_ layered	10.937	2743.5	0.0488	5.372	0.0951
bare_ midi	effects_ layered	6.533	211.8	0.0449	5.414	0.0672

Metadata columns (sr_hz=44100, n_samples_compared=1_323_000, embedding_rung=vggish) are constant across pairs. The three families **disagree** on which of {bare_midi, effects_layered} is closer to original: envelope + mel-L1 rank bare_midi closer; spectral-centroid is essentially tied; VGGish cosine inverts and ranks effects_layered closer. This is not measurement noise — the LUFs-M gap is 2.68 vs 5.37 LU and the embedding gap 0.095 vs 0.123 is consistent — and it is exactly the informative disagreement the M-TEX-1/panel <mechanism> block predicted: envelope and mel-L1 measure things the effects chain damages by design (mean band energy, temporal loudness), and the perceptual embedding measures the auditory-scene features fluidsynth’s dry, close-mic’d mix lacks and that reverb + chorus supply. The seed-fallback ladder was walked in order: seed_mid_50s and seed_long_87s were rejected on spectral evidence (both are 220 Hz sinusoidal test tones — FFT peaks at ~220 Hz with the next five bins $\leq$ 660 and peak/RMS ratios characteristic of pure sines), so rung (c) synth_030s was chosen with the verdict downgraded to /medium and the weaker “bare-MIDI-vs-fluidsynth-mix gap” claim substituted for the stronger “bare-MIDI-vs-recorded-original gap” claim throughout. The auditor re-ran the pipeline from a fresh directory and reproduced all four SHA-256 prefixes exactly (153997a829f2b42c / fc8c3ecbfff073d2 / 13d7238637d1ee31 for the three WAVs; b3570a795c8c3e7a for the TSV). Defence-in-depth against aggregation is at three layers (panel PUBLIC_KEYS assert, panel _BANNED_KEYS sweep, measure_across_stages.py’s OWN BANNED_AGGREGATE_KEYS sweep). Cross-branch integration test §19 (24 checks including the four SHA-256 baselines) is green.

Campaign-level state at cycle-9 exit

- **Done in this range:** M-SCORE-1 (/high); M-RULES-1 (/high on both halves — schema at cycle 6, extraction at cycle 9); M-TEX-1 (/medium on both halves — panel at cycle 4, stage-by-stage at cycle 9); M-INGEST-1/egress-ready-automation (/high); M-TRANS-1/basic-pitch/octave-suppression (/high invalidated, negative finding closed).
- **Ledger + plan:** ledger grew by 10 events on cycle 9’s clone 0 alone; the plan-of-record now carries five-column rows for every sub-milestone emitted by cycles 8-9. Every event uses the healthy schema (ts / narrative / nested confidence / explicit event_id / agent / cycle) — the cycle-7 lesson (missing event_id on four rows, repaired in-cycle) has stuck.
- **Environment:** music21 9.1.0 added at cycle 8 without touching numpy 1.26.4, TF 2.21, the basic-pitch venv, or the DawDreamer / Surge XT chain. No further drift in cycle 9.
- **Blocked on rated audio:** parent M-EAR-1 v0 training; the state machine that will unblock it is on disk, IDLE, and awaits two consecutive fresh media_ok=true rows.

Discussion

Three things about this range are worth naming.

First, the pattern of closing the chassis half of a milestone before the data half arrives — established in cycles 4-6 with the M-EAR-1 preparation branch — reached full effect this range.

M-SCORE-1's bridge, M-RULES-1's extraction, M-TEX-1's stage-by-stage measurement, and the egress-ready state machine are all now unblockable-into-final-form: when the rated bytes appear, no new engineering is required to route them through classification, chunking, and rating training. The state machine is the last piece of that pattern: it turns the egress delay from a blocker into a scheduling constraint that the campaign can wait out without human polling and without downstream code paths that must be added later.

Second, the escape-hatch discipline held across two very different kinds of negative finding. The octave-suppression branch published the full 3×3 grid, diagnosed the shortfall at the level of individual pitches, and filed the sub-milestone as *invalidated/high* — a negative finding that leaves the ceiling on this algorithm family known for downstream consumers rather than smearing it with a passing number obtained by re-tuning. The stage-by-stage branch rejected two seed-ladder rungs on spectral evidence, fell to the third with a verdict downgrade, and substituted a weaker claim throughout the report so the escape-hatch consequences carried through rather than being buried under an unchanged headline. Both are worth preserving as canonical examples of how a falsifiable claim is supposed to fail without becoming an untraceable pass.

Third, the family-disagreement finding on M-TEX-1/stage-by-stage is the panel design's first-contact validation. The panel was built on the commitment that different families of measure measure different things and their disagreement is signal, and the eight-key contract with a triple-layer aggregation ban is that commitment made mechanical. The synth-mix seed produced exactly the kind of informative disagreement the mechanism block predicted — envelope and mel-L1 penalise the effects chain while VGGish rewards it — and the report refuses to reduce that trade to a scalar. When rated audio arrives, the same pipeline can be re-run to test whether the pattern holds on non-synthetic sources; that is a natural falsifiability check the current branch enables but does not consume.

The rules-ledger side of Goal G4 is now the most-complete goal in the campaign after G1. The 28 seed rules are a live example, not a theoretical schema, and the round-trip through `read_ledger()` / `effective_rules()` is exercised end-to-end. The known limitations on rhythmic-v1 and form-v1 are recorded as future-supersede cases with concrete triggers (a real drums transcription; a longer seed with detectable measure-scale structure) rather than as defects.

Open Questions

- **First deterministic generation from the rules ledger (M-GEN-1).** Now unblocked; needs a target harmonic / rhythmic / melodic / form / arrangement selection over `effective_rules()` and a rendering path back to audio via `fluidsynth-on-merged-MIDI`. Deterministic content-addressed rule ids and the append-only ledger discipline are the load-bearing invariants any generator must respect.
- **M-EAR-1 v0 training** — still blocked on rated-audio egress. The state machine will fire the pipeline unattended when two consecutive fresh `media_ok=true` rows land in the probe log; the M-EAR-1 chassis (features, CORN head, leak-test) is ready for immediate reuse.
- **Fixed-point iteration on octave suppression.** One-line wrapper over the existing filter and grid; expected $\approx +0.10$ additional aggregate bass F1 uplift based on the pitch-set arithmetic. Not prioritised above M-GEN-1.
- **rhythmic-v2** once a non-empty drums transcription lands (either basic-pitch re-run on a longer seed, or the alternative drums transcriber the M-TRANS-1 report recommended) — supersedes the fallback-labelled "kick" rows.
- **form-v2** — novelty-curve boundary detection on a ≥ 60 s seed where measure-scale structure is genuinely detectable; supersedes the five-parallel-sectionization strategy. If it turns out to be needed, a schema tweak allowing a strategy field on form rows (detected

vs uniform-N vs multi-strategy) is the right accompanying change.

- **M-SCORE-1 refinement.** Investigate why merged_synth030s.musicxml reports 131 nominal measures for 30 s of content; if fixable, trailing-measure padding should be trimmed so downstream end_measure values carry musically meaningful semantics.
- **M-TEX-1 to /high.** Independent unblocks: real recorded audio (blocked on egress-ready-automation firing) and/or the CLAP-rung swap on M-TEX-1/panel/embedding.
- **CLAP-rung swap on the texture panel's embedding.** A future cycle can revise the family-disagreement pattern under a CLAP-trained perceptual embedding; whether the pattern holds under a different perceptual model is a natural falsifiability check.
- **Append-helper event_id default.** The cycle-7 lesson (append helper accepts events without event_id) has now been avoided twice by explicit-event_id discipline in cycles 8 and 9. A defensive default or a docstring update remains queued.

Appendix: Provenance

Cycle range: cycles 7-9. **Working directory:** /home/user/long-exposure-runs/music-gen.

Session references: cycle 7 researcher 7ef32dd0-70b4-44c4-8c9d-1b2210ffb068; cycle 8 worker 6c80005c-53bb-40bf-beb2-cba0223803f9; cycle 9 researcher 00a87607-73af-437b-bcf4-f8569fa27ca7.

Sub-agent transcripts (fork clones).

- Fork 3a908edcb241 (cycle 8): clone 0 (M-SCORE-1) — researcher 843cb35b-9232-47b4-925c-a94c8d4ae257, worker ef72c18a-a76d-406c-87ae-f8243b0ba861, auditor 0d95720c-be8d-4925-a72e-2f464664d68b. Clone 1 (M-TRANS-1/basic-pitch/octave-suppression) — researcher 6f1cfefb-0c92-45eb-8eab-50e5c0f26095, worker af5ea339-1b1b-4a47-8309-045a70599ad5, auditor 879f06da-0072-49a7-849f-2d04c6f8b34c. Clone 2 (M-INGEST-1/egress-ready-automation) — cycle 1 researcher 8c17d07b-5cb3-40d7-b0d7-bd44e5cb72fb, worker fd0c6c9e-34bc-4b0d-b3ea-04c775716b71, auditor fffd9a03-d17b-48e9-a350-9ab90c4cdd95; cycle 2 researcher 6853eb97-6044-4369-b334-52767b466268, worker 925bafba-2111-49cd-80f8-d05134482317, auditor 2b66a842-d318-415f-bb9a-389508e1bb54.
- Fork f1bae241bde9 (cycle 9): clone 0 (M-RULES-1 extraction) — researcher d797a336-45d8-4451-ac91-325ab2e2c768, worker 2803dfd0-ddfa-4816-b5ef-f620d52f941b, auditor 0250497a-5491-442d-b56f-8cecaf57752f. Clone 1 (M-TEX-1 stage-by-stage) — researcher 44aa9e94-e4af-40d1-adeb-33c7a46bc92c, worker ed5a862a-6694-47b3-be68-49c612056085, auditor 06901b63-38db-480d-9320-33c634326183.

Branch-scoped reports on disk (all authored in this range):

- docs/score_bridge_report.md — 304 lines; round-trip proof, merged full-song F1 table, API reference, failure modes, reproducibility, non-factor isolation, determinism scrubbing list.
- docs/basic_pitch_octave_refinement.md — 246 lines; the negative-finding report with the full 3×3 grid, pitch-level diagnostic, and follow-up recommendations.
- docs/egress_ready_automation.md — 305 lines / 15 036 bytes; state diagram (mermaid), trigger rule with falsifiability criteria, six-scenario matrix, state persistence with worked crash-between-CHUNKING-and-CLASSIFYING example, failure recovery, human-override API, isolation, handoff, reproduction.
- docs/rules_extraction_report.md — 231 lines; five extractor designs, coverage figure, verification matrix, representative rule content, provenance resolvability, honest limitations.
- docs/tex_stage_by_stage_report.md — 309 lines; seed-selection ladder walk, 24-number panel table, per-family commentary, family-disagreement interpretation, determinism proof, artefact SHAs.

Reproduction of load-bearing SHAs at cycle-9 exit:

d87fa0f5e6d87e6be551fcfb4e844a35c247733c42b19452d416b5ba573b0ec2 data/transcribe/octave_suppression/grid_search/4fe722adde034c099ff9e65437f0d5c138cb3dd2595089960150af5c2546fc4b data/rules/ledger.jsonl
153997a829f2b42c... data/tex/renderers/synth_030s/original.wav
fc8c3eccbff073d2... data/tex/renderers/synth_030s/bare_midi.wav
13d7238637d1ee31... data/tex/renderers/synth_030s/effects_layered.wav
b3570a795c8c3e7a... data/tex/stage_by_stage_synth_030s.tsv

Tests at cycle-9 exit. tests/test_score_bridge.py — 23/23. tests/test_octave_suppression.py — 14/14. tests/test_egress_ready_state.py — 62/62. tests/test_rules_schema.py — 25/25. tests/test_rules_extraction.py — 34/34. tests/test_integration_cross_branch.py — 0 failures across §14 (octave suppression, 27 checks), §15 (score bridge, extended in cycle 8), §17 (egress-ready, 52 checks), §18 (rules extraction, 33 checks), §19 (tex stage-by-stage, 24 checks including four SHA-256 baselines), on top of the prior §§ carried forward.

Environment stack unchanged since cycle 8: mscore3 MuseScore3 3.2.3 (headless, QT_QPA_PLATFORM=offscreen); Python 3.11.15 (/usr/bin/python3); numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; mido (Debian package); fluidsynth (Debian package) with pinned SF2; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/. Single-thread BLAS pins everywhere numeric determinism is required.

Handoff to next cycle. The natural next research step is M-GEN-1 — first deterministic generation from the 28-row rules ledger — which is now unblocked end-to-end. Rated-audio arrival remains the M-EAR-1 unblock; the state machine will handle it unattended and the M-EAR-1 preparation chassis is ready for immediate reuse. The follow-ups listed under **Open Questions** are queued in priority order.